INCH-POUND

MIL-C-24643/52D
22 November 1994
SUPERSEDING
MIL-C-24643/52C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTPNW (INCLUDING VARIATION LSTPNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSTPNW (unarmored) and LSTPNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, AWG size 22, class K stranding.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation or ethylene propylene rubber insulation., 0.009 inch minimum average thickness. Telephone identification code shall be applied by method 3.
Fourth - Two conductors cabled together to form a pair. Maximum lay 2 inches.
Fifth - The required number of pairs (see table I), cabled with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Sixth - Binder tape applied helically with overlap.

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Construction (non-watertight) (continued)

Construction variant for type LSTPNW

Seventh- Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant for type LSTPNWA

Seventh- Cross-linked polyolefin jacket. (see table I for thickness).
Eighth - Braided metal armor.

TABLE I. Details

Military 1/ part no. M24643/52	Size 2/	Number of pairs	Cable jacket thickness min avg (inch)	Overall diameter (inches) TYPE LSTPNW min	TYPE LSTPNW max	TYPE LSTPNWA min	TYPE LSTPNWA max	Cold bending mandrel diameter (max inch)
-01UN	1-1/2	1-1/2 3/	0.040	0.217	0.235	0.267	0.285	3.0
-02UN	3	3	.040	.287	.310	.337	.360	3.5
-03UN	5	5	.040	.338	.365	.388	.415	4.0
-04UN	10	10	.040	.435	.470	.485	.520	5.5
-05UN	15	15	.040	.490	.530	.540	.580	6.5
-06UN	20	20	.040	.532	.575	.582	.625	7.0
-07UN	30	30	.050	.629	.680	.679	.730	8.0
-08UN	40	40	.050	.708	.765	.758	.815	9.0

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSTPNW and LSTPNWA.
* 3/ Size 1-1/2 is composed of three conductors cabled to form a triad.Color coded black, white and red.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum.	17.94
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulation resistance - Conductor to conductor, Megohms/1000 feet, minimum	1000
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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EXAMINATION AND TESTS (continued):	Requirements
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	25
Armor - conformance to material, construction and coverage (type LSTPNWA only)	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTPNW-1-1/2 through -10	2500
LSTPNW-15 and -20	1500
LSTPNW-30 and -40	1000

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